DRUG MICROBIOTA PROJECT

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Introduction

This dataset describes how different drugs interact with gut bacteria (microbiota) and how these interactions may affect the drug's metabolism, effectiveness, stability, and behavior inside the body, it connects drugs, bacteria, metabolism and drug performance, it shows how drugs are metabolized, absorbed, activated, or deactivated inside the body. This dataset belongs to the domain of pharmacology, drug microbiota interactions, and precision medicine.

Data Source Kaggle: https:

www.kaggle.com/datasets/ziya07/ai-driven-drug-stability-prediction-dataset

Key Variables

Variable	Type	Notes
Drug_Name	Categorical (Nominal)	-These are names of medicines (e.g., Ibuprofen, Atorvastatin). -Names cannot be ordered (Ibuprofen is not "higher" or "lower" than Metformin). -Name of the medicine being studied
Drug_Class	Categorical (Nominal)	-Classes like PPI, NSAID, Statin are categories, not numbers. -There is no ranking between classes (Statins are not "greater" than PPIs). -the category the drug belongs to
Bacteria_Species	Categorical (Nominal)	-Species names (E. coli, Klebsiella pneumoniae, Enterococcus faecalis) are labels. -They cannot be ordered meaningfully. -The specific type of the bacteria that interacts with the drug
Enzyme	Categorical (Nominal)	-Enzyme types like Protease, Nitroreductase, Azoreductase are names, not values.

		-the type of enzyme produced by the bacteria that affects the drug
Metabolisim_Pathway	Categorical (Nominal)	-Values like Hepatic, Renal, Mixed are groups, not numbers. - The route the drug takes when being broken down in the body. Examples: liver, kidney, gut, or mixed.
Solubility_Ph	Numerical	- pH is measured on a continuous numeric scale. -How well the drug dissolves into the body
Bioavailability	Numerical	-0 means “no drug is absorbed,” which is a true zero ratio scale. -How much of the drug acatually gets absorbed into the bloodstream
Half_Life_hrs	Numerical	- Measured in hours, a continuous scale. -How long it takes for half of the drug to leave the body
Plasma_protein_binding	Numerical	-Given as decimal fractions (e.g., 0.78). -How much of the drug sticks to blood proteins. -Exists on a continuous numeric scale.
Excreation	Categorical (Nominal)	-Routes like Renal, Hepatic, Biliary are categories. - How the drug leaves the body.
Depletion_Probability	Numerical	-It represents probability from 0 to 1. -How likely bacteria can destroy or reduce the drug
Altered_Efficacy	Ordinal	-Only two values exist, 0 and 1. -0 = no change -1 = drug effect altered - Whether the drug becomes less effective after interacting with bacteria.
Stability_Score	Numerical(Continous)	- The difference between scores is meaningful.

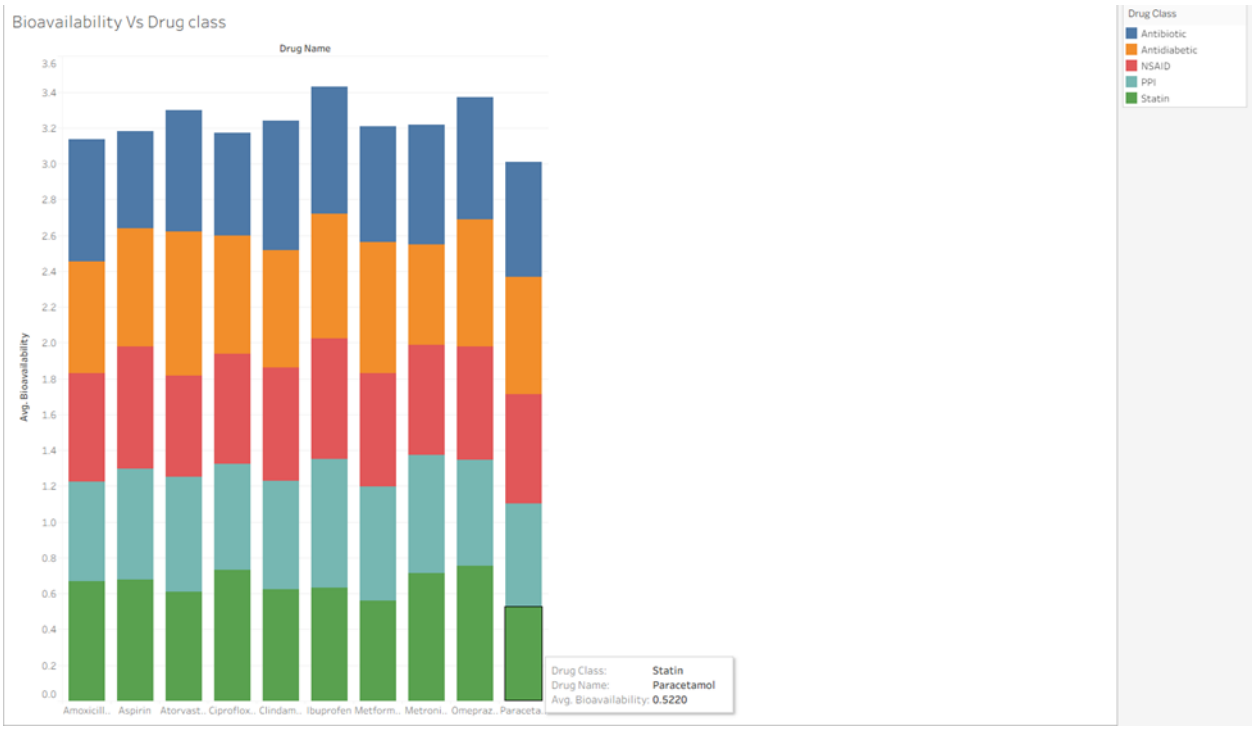
		- How stable the drug is when exposed to bacteria and enzymes.

OBJECTIVES AND GOALS:

- Identify overall patterns and trends in drug microbiota interactions to understand which drugs and bacteria are most commonly involved in metabolic modification.
- Compare drug categories (e.g., NSAIDs, PPIs, Statins) to determine how different classes vary in bioavailability, stability, degradation probability, and likelihood of altered efficacy.
- Analyze relationships and correlations between drug properties (solubility, half-life, protein binding, bioavailability) and microbiota influence indicators (depletion probability, stability score, altered efficacy).
- Determine the impact of bacteria and their enzymes by identifying which bacterial species and enzyme types are most strongly associated with drug breakdown or reduced effectiveness.
- Evaluate drug stability and degradation patterns to identify drugs that are more vulnerable to microbiota induced modification.
- Assess the role of metabolism pathways and excretion routes to see how biological processing (hepatic, renal, mixed) influences drug microbiota interactions.
- Identify high risk drug bacteria combinations that show elevated depletion probabilities or consistent alterations in drug efficacy.

PERFOMED TASKS

1) BIOAVAILABILITY VS DRUG CLASS



This stacked bar chart shows how different drug classes contribute to the total bioavailability of each drug. Each bar represents one drug class, each colored section inside the bar represents one drug in that class. The height of the whole bar shows the total bioavailability in that class.

Reason:

- It allows the comparison of the drug classes across different drugs
- Shows the distribution of bioavailability within each drug
- Effective on showing composition patterns

Key Observations:

- Overall bioavailability levels are fairly similar across most drugs.
- Antibiotics contribute the highest portion of bioavailability in most drugs.
- Antidiabetic drugs also contribute strongly to bioavailability.

Marks

Mark type: Bars which represents a drug and each segment within the bar represents the bioavailability contributed by each drug class.

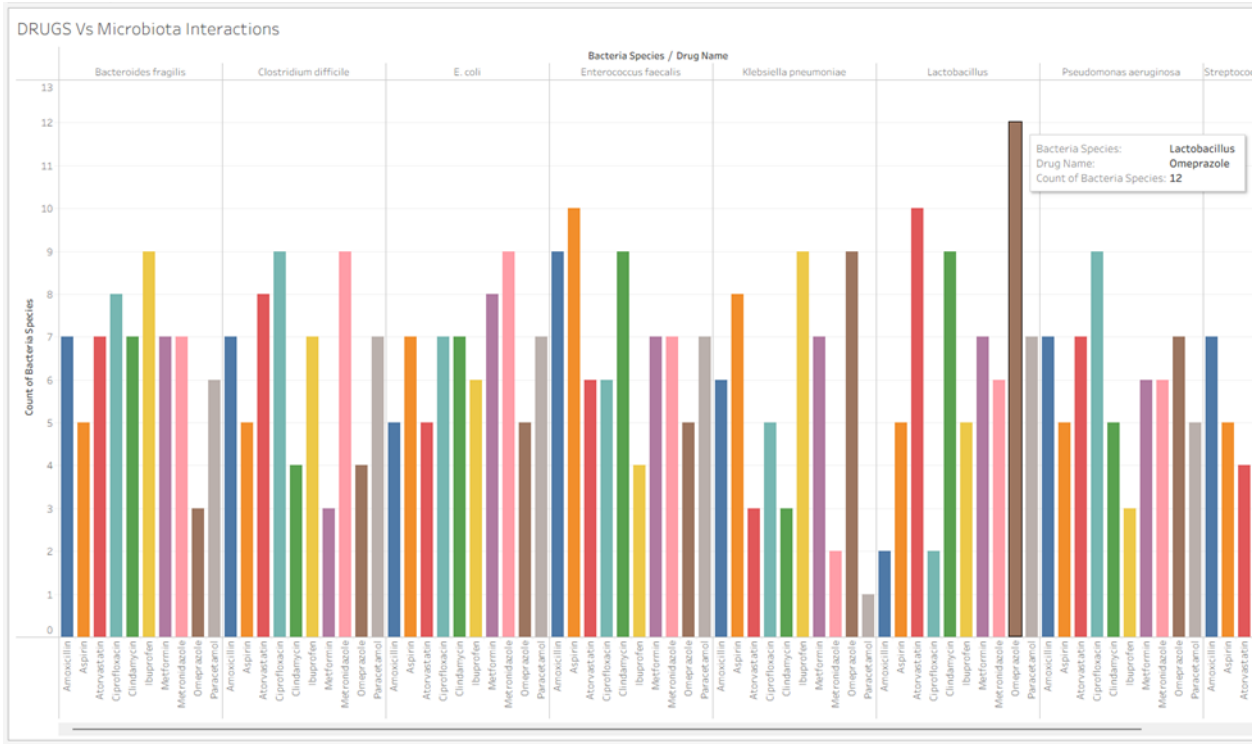
Channels and Encoding:

- ✓ Color hue - used to differentiate Drug Classes (e.g., Antibiotic, Antidiabetic, NSAID, PPI, Statin).
- ✓ Size - The height of each bar segment (size along the y-axis) encodes bioavailability value.
- ✓ Positions x and y axis
- ✓ Labeling

Aesthetics

- Legend
- Sorting
- Axes formatting
- White background
- Subtle gridlines

2) DRUGS VS MICROBIOTA INTERACTIONS



This is a Grouped Bar Chart, in each group represents a bacterial species (e.g., E. coli, Lactobacillus), within each group, bars represent different drug names (Amoxicillin, Aspirin, Ibuprofen, etc.), the height of each bar shows the count of interactions between a specific drug and that bacterial species. This chart allows comparison both within groups (across drugs) and between groups (across bacterial species).

Reason:

- Comparing Categories across multiple Groups
- Supports detailed and high granularity
- Helps identify patterns and trends

Key Observations:

- Which drugs are heavily influenced by microbiota
- Which bacteria interact with many drugs

- ❖ Patterns of drug–bacteria relationships
- ❖ Drugs that are more “microbiota-sensitive”

Marks

Mark Type: Bar, the visualization uses bars as the primary mark.
Each bar represents the count of interactions between one drug and one bacterial species.

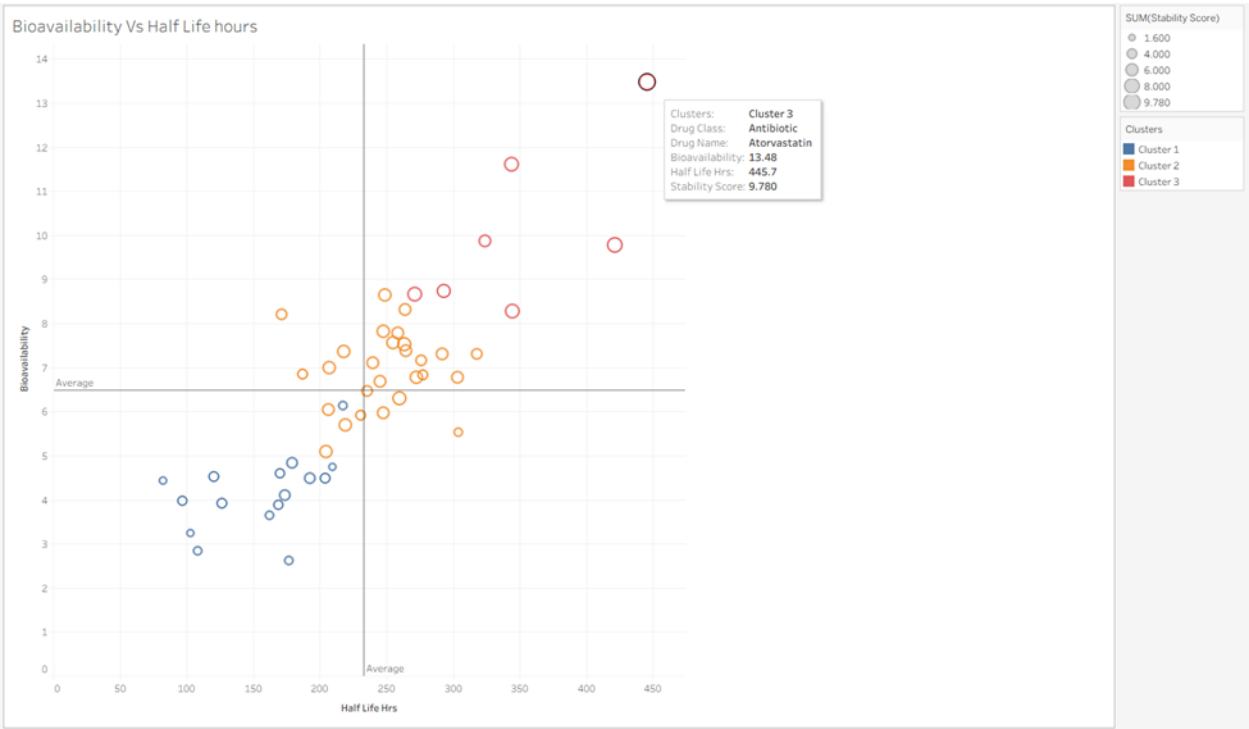
Channels and Encoding

- ✓ Color hue(categorical) is used to differentiate Drug names, each drug has a distinct color.
- ✓ Size - the height of each bar represents the count of interactions; taller bars indicate higher bacterial drug interaction frequencies.
- ✓ Positions - x and y axis
- ✓ Labeling - x-axis labels show bacterial species and drug names; y-axis labels show interaction counts.

Aesthetics

- Legends
- Sorting
- Axes Formatting
- Titles and subtitles
- Layout choices
- White background
- Subtle gridlines

3) BIOAVAILABILITY VS HALF LIFE HOURS



This Scatter Plot compares each drug's bioavailability and half-life to show how well the drug is absorbed and how long it stays active in the body. The points are grouped into three clusters based on drug stability and behavior.

Reason:

- Relationships between Two numeric values
- Allows Multivariate Encoding (Color and size)
- Effective for outlier detection

Key Observations:

- High performing drugs have high bioavailabity and long half-life
- Low bioavailability drugs also have short half-life

- ❖ Stability score increases with bioavailability and half-life

Marks

Mark Type: Point, the visualization uses points (circles) as the primary mark.
Each point represents a drug in the dataset.

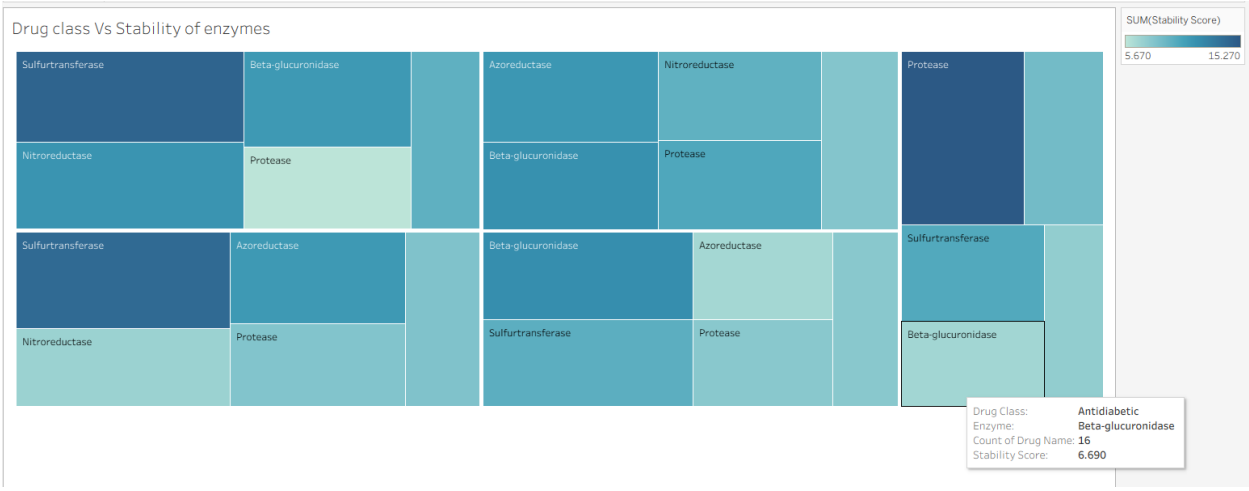
Channels and Encoding

- ✓ Color hue is used to represent Cluster Groups (Cluster 1, Cluster 2, Cluster 3), Each cluster is assigned a distinct color.
- ✓ Size - the size of each point encodes the Stability Score, Larger circles = higher stability; smaller circles = lower stability.
- ✓ Reference Lines - horizontal average line for bioavailability, vertical average line for Half-Life
- ✓ Opacity - circles help reduce overlap in dense areas.
- ✓ Positions - x and y axis
- ✓ Labeling - Tooltips show: Cluster, Drug Class, Drug Name, Bioavailability, Half-life, Stability score, Axis labels: Half Life Hrs, Bioavailability

Aesthetics:

- Legends
- Sorting
- Axes Formating
- Titles and subtitles
- Layout choices
- White background
- Subtle gridlines

4) DRUG CLASS VS STABILITY OF ENZYMES



TreeMap

This treemap shows how stable bacterial enzymes are when interacting with drugs from different drug classes. Larger and darker tiles represent enzyme drug combinations with higher stability, meaning the microbiota can strongly influence or resist drug breakdown

Reason:

- Works well for Dense or Categorical data
- Helps identify patterns quickly
- Encodes two variables at once

Key Observations:

- Beta-glucuronidase appears frequently and with medium-to-high stability
- Protease has lower and more variable stability

- ❖ Different drug classes interact with enzymes at different stability levels
- ❖ High-stability enzymes may reduce drug effectiveness

Marks

Mark Type: Area (Rectangles)

This visualization uses rectangular areas as the primary mark.
Each rectangle represents an enzyme, and its size corresponds to its stability or count.

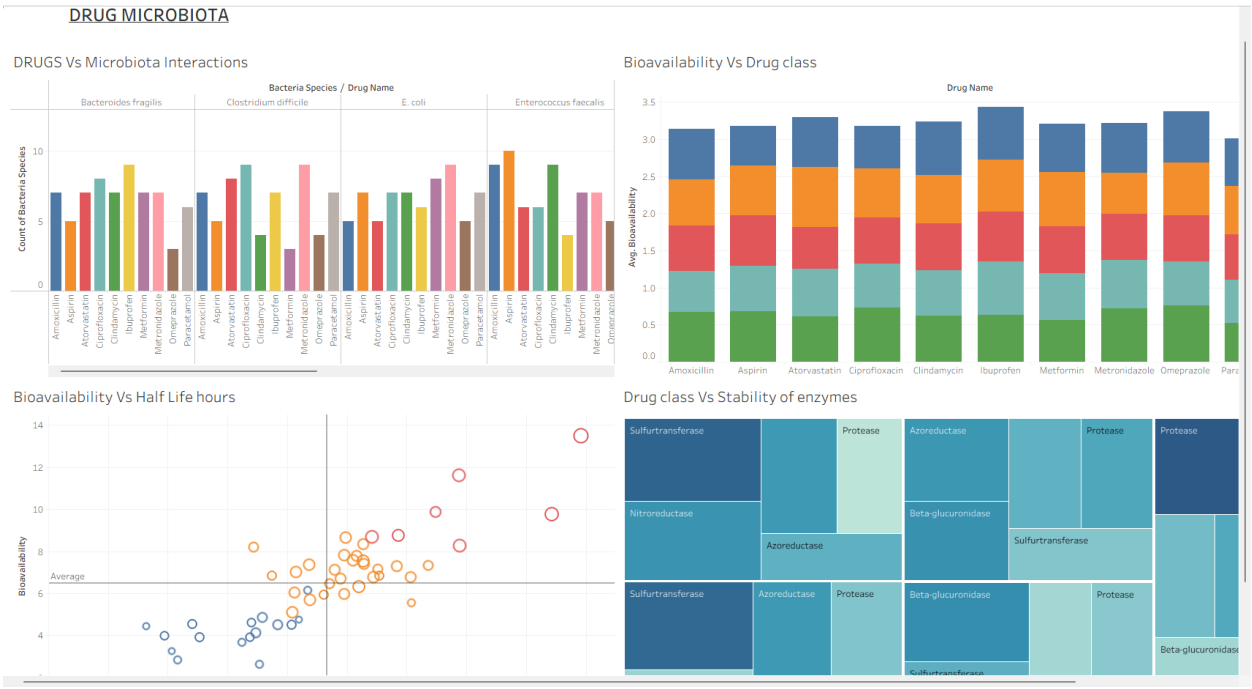
Channels and Encoding

- ✓ Color Luminance, uses a sequential color scale that ranges from: dark teal is the higher stability score, light teal is lower stability score
- ✓ Size
- ✓ Area (rectangle size) represents the sum of stability score or count of drug interactions, larger rectangles represent higher score or more interactions.
- ✓ Position is used to group enzymes within the treemap layout, Items are placed adjacent to each other within a larger rectangle (representing the drug class grouping or category level).
- ✓ Labeling each rectangle is labeled with the enzyme name.
- ✓

Aesthetics:

- Legends
- Sorting
- Axes Formating
- Titles and subtitles
- Layout choices
- Soft boarders

DASHBOARD



Dashboard Layout

The dashboard layout is designed to guide the viewer through a logical analytical flow starting with overall drug microbiota interactions (What bacteria does the drug affect), comparing drug class performance(How well the drug is absorbed), analyzing quantitative relationships(how long does the drug stay alive), and finally examining enzyme stability patterns(which enzymes process the drug). Visuals are placed to maximize readability, support comparison, and maintain balance, ultimately helping the user understand the full picture of drug microbiota interactions.

How the dashboard solves the Objectives

Visual	Objectives
Drugs Vs Microbiota Interactions	Identifying patterns and trends,

Bioavailability Vs Drug class	Comparing drug categories
Bioavailabity Vs HalfLife	Analyzing relationships between drug properties
Drug Class Vs Stability of Enzymes	Determining the impact of bacterial species

Insight:

By interacting with the dashboard, users gain insights into drug microbiota interactions, bioavailability differences across classes, relationships between half-life and absorption, and enzyme stability patterns. The combined visuals allow users to explore trends, compare categories, detect correlations, identify outliers, and understand biochemical stability mechanisms ultimately supporting informed analysis and decision-making.

Conclusion:

This dashboard highlights how different drugs interact with key gut bacteria and how these interactions influence drug behavior. The visuals show that some bacteria, like E. coli and Bacteroides fragilis, interact with many drugs, while bioavailability and half-life vary widely across drug classes. The scatterplot reveals that drugs with higher stability often have higher bioavailability and longer half-lives. The enzyme treemap further emphasizes that enzymes such as Sulfurtransferase and Beta-glucuronidase have strong, stable interactions across drug classes. Overall, the dashboard clearly demonstrates the important role microbiota plays in shaping drug absorption, stability, and effectiveness.